MONTEFIORE MEDICAL CENTER
DEPARTMENT OF CLINICAL MICROBIOLOGY
111 EAST 210TH STREET
BRONX, NY 10467

STANDARD OPERATION PROCEDURE NUMBER 15
MALDI BIOTYPER CA SYSTEM

A. Background

The MALDI Biotyper (MBT) CA system is an FDA-approved instrument for the identification of bacteria and yeast by MALDI (matrix-assisted laser desorption/ionization)-TOF (time of flight) mass spectrometry. Identification of microorganisms is based on protein expression profiles. MALDI identification takes only minutes and can be achieved with as little as one bacterial or yeast colony. In contrast, biochemical testing can take 4-24 hours and may require subculture, further delaying turnaround time. Thus, the use of MALDI can significantly decrease the time to organism identification in the microbiology laboratory.

B. Test Principle

MALDI-TOF analysis can be performed from isolated colonies or directly from positive blood culture bottles using the MBT Sepsityper Kit US IVD.

<u>Colony Identification</u>

An isolated colony or protein extraction supernatant is transferred to a selected position on a target plate (US IVD MBT Biotarget 96 US IVD). Organisms to be identified should be isolated for purity on appropriate isolation media and after the indicated amount of timed growth (see Procedure below). The MALDI process transforms the proteins and peptides from the isolated microorganisms into positively charged ions. This is achieved by irradiating a matrix-sample composite with a UV laser. The matrix absorbs laser energy and transfers protons to the intact proteins or peptides in the gas phase. These ions are electrostatically accelerated and arrive in the flight tube at a mass-dependent speed. The MALDI Biotyper CA System measures the time between pulsed acceleration and the corresponding detector signal of the ions, and the time is converted into an exact molecular mass. A characteristic mass and intensity distribution pattern, consisting primarily of the highly abundant ribosomal proteins, is species-specific for many bacteria and yeasts and can be used as a 'molecular fingerprint' to identify a test organism. The spectra of the unknown organisms is compared to a reference library and using a statistical algorithm, a log(score) value between 0.00 and 3.00 is calculated. The higher the log(score) value, the higher the degree of similarity to a given organism in the reference library. A log(score) value ≥2.00 can be considered as a high-confidence identification of the test organism.

<u>Positive Blood Cultures</u>

The MBT Sepsityper Kit US IVD is a disposable blood culture processing device that includes associated reagents that are intended to concentrate and purify microbial cells from blood culture samples identified as positive by a continuous monitoring blood culture system and confirmed to demonstrate the presence of a single organism as determined by Gram stain.

C. Reagents, Instruments, Supplies

- MALDI Biotyper CA System
- MBT Galaxy matrix/formic acid spotter (Note: targets may be prepared by manually pipetting reagents or using the Galaxy instrument. See Galaxy SOP for more details).
- MBT Biotarget 96 US IVD (pack of 20 target plates) [P/N: 1840380]
- MSP Biotarget Adapter [P/N: 8267615]
- US IVD BTS [P/N: 8604530]
    - Store at ≤ -18°C
- US IVD HCCA portioned [P/N: 8604531] or US IVD Galaxy HCCA (1852132)
    - Store 2-8°C

    - HCCA can cause skin, eye, and respiratory tract irritation. See SDS for further information.
- Bruker Standard Solvent (acetonitrile 50%, water 47.5% and trifluoroacetic acid 2.5%) [Vendor: Sigma-Aldrich or equivalent]
    - Causes skin irritation and serious eye damage and may cause respiratory damage if inhaled. See SDS for further information.
- HPLC-grade water
- Acetonitrile
    - Store in fire-proof chemical safety cabinet. Liquid and vapor are flammable. Can cause mild skin irritation and serious eye damage. Check reagent expiration before use. See MSDS for further information.
- Formic acid
    - Causes skin burns and eye damage and is toxic if inhaled. (See SDS for further information.)
- Absolute ethanol
- Sterile Colony Transfer Device
- Pipette tips 0.5-20 µL, 2-200 µL, 50-1000 µL [Biopure or PCR grade]
- Clear plastic tubes,1.5 mL and 0.5 ml [Vendor: Eppendorf preferred]
- Bench-top microcentrifuge capable of 13,000 to 15,000 rpm
- Blood culture vent
- 3ml syringe with needle
- MBT Sepsityper Kit US IVD
    - Store 15-25°C
    - Keep kit components out of direct sunlight
    - **Kit is stable for up to 3 months after opening**

D. Solution Preparation

70% Formic Acid: Add 300µl of HPLC-grade water to a microcentrifuge tube. Then add 700µl of formic acid. 70% formic acid is stable at room-temperature for 7 days. If the solution is used in the MBT Galaxy spotter, working solution is stable for 30 days due to the controlled environment and larger volume (1.5ml water, 3.5 ml formic acid) is prepared. **NOTE: Working solution tubes that contain 70% formic acid will be labeled as "FA."**

1. 100% Acetonitrile: Transfer to a 1.5 ml microcentrifuge tube. **NOTE: Label working solution as "A."**

HCCA matrix IVD (red top): Reconstitute the vial by adding **250µl of Standard Solvent**. Vortex, ensuring that there is no precipitate before use. Reconstituted matrix is stable at 20-25°C for 7 days.

HCCA matrix Galaxy (green top): Reconstitute the vial by adding **750µl of Standard Solvent**. Vortex, ensuring that there is no precipitate before use. Reconstituted matrix is stable at 20-25°C for 7 days.

4. BTS: Reconstituted BTS is stable at -18°C or below for 5 months or 24 hours at 20-25°C.
    - Bring US IVD BTS to room temperature for a minimum of five (5) minutes before use.
    - Add 50 µL of Standard Solvent and dissolve by pipetting up and down at least 20 times.
    - Incubate the US IVD BTS solution for five (5) minutes at room temperature and repeat mixing by pipetting up and down at least 20 times.
    - Centrifuge at 13,000 rpm for 2 minutes at room temperature.
    - Aliquot into 0.5ml microcentrifuge tubes for storage at -18°C or below.
    - Complete the BTS preparation log.

E. Environmental Requirements

- 20-25°C – Test organism sample preparation
- 16-33°C - Instrument operation

F. Safety Precautions

- Use PPE when handling all reagents and specimens.
- CDC pathogens of concern/ bioterrorism agents must be ruled out before MALDI is performed.
- Formic acid stock bottle should be opened in a fume hood. Small volumes of 70% formic acid may be used on the open bench per NYS Wadsworth Center guidance.
- SDS for all reagents are available on the Montefiore Intranet.
- The MALDI CA System contains a laser. Do not open the instrument to avoid harm from the laser.

G. Procedure for Sample Preparation – Extended Direct Transfer

1. Using a sterile colony transfer device, smear an isolated colony of bacteria or yeast as a thin film directly onto a sample position on a MBT Biotarget 96 US IVD. **For bacteria, spot a duplicate well with the same or new colony with similar morphology. For yeast, spot in triplicate.**

Note: After incubation of bacteria on **recommended isolation media** for 18-48 hours at (34-37°C) or incubation of yeasts on recommended isolation media for 18-48 hours at (29°C±2°C), colonies are stable for up to 12 hours when held at room temperature.

Specific requirements:

- *Bordetella*: Cultivation on BG agar should not be longer than 24 hours (+12 hours storage at RT).
- *Campylobacter*: Cultivation can be prolonged to 72 hours (+12 hours storage at RT).
- *Streptococcus pneumoniae*: Cultivation should not be longer than 24 hours (+12 hours storage at RT) due to possible autolysis.
- *Neisseria*: Cultivation on Modified Thayer-Martin Agar should not be longer than 24 hours (+12 hours storage at RT).
- *Nocardia*: Identification rate may decrease after incubation time >48h (+12h storage at RT).

Recommended Media

- Columbia blood agar with 5% sheep blood (Gram-negative bacteria)
- Trypticase soy agar with 5% sheep blood (Gram-negative bacteria,Gram-positive bacteria, yeasts)
- Chocolate agar (Gram-negative bacteria, Gram-positive bacteria)
- MacConkey Agar (Gram-negative bacteria)
- Columbia CNA agar with 5% sheep blood (Gram-positive bacteria)
- Brucella Agar with 5% horse blood (Gram-negative anaerobic bacteria, Gram-positive anaerobic bacteria)
- CDC anaerobe Agar with 5% sheep blood (Gram-negative anaerobic bacteria, Gram- positive anaerobic bacteria)
- CDC anaerobe 5% sheep blood Agar with phenylethyl alcohol (Gram-negative anaerobic bacteria, Gram-positive anaerobic bacteria)
- CDC anaerobe laked sheep blood Agar with kanamycin and vancomycin (Gram- negative anaerobic bacteria)
- Bacteroides bile esculin Agar with amikacin *(Bacteroides* species)
- Clostridium difficile Agar with 7% sheep blood (*Clostridium difficile*)
- Sabouraud-Dextrose Agar (Yeasts)
- Brain Heart Infusion Agar (Yeasts)
- Campylobacter Agar with 5 Antimicrobics and 10% Sheep Blood (*Campylobacter* species)
- Bordet Gengou Agar with 15% sheep blood (*Bordetella* species)
- Buffered Charcoal Yeast Extract Agar (*Legionella* species)
- Buffered Charcoal Yeast Extract Selective Agar with polymyxin, anisomycin and vancomycin (*Nocardia* species)
- Modified Thayer-Martin Agar (*Neisseria* species)

2. Overlay the sample spot with 1 µL 70% formic acid.

3. Add 1 µL of reconstituted US IVD BTS onto **2 target wells** and dry the target at room temperature.

4. Overlay each of the sample positions and BTS control positions with 1 µL US IVD HCCA portioned solution. **Each run must also include 1 well with HCCA but no organism as a negative control.** Dry the inoculated target at room temperature.

Note: US IVD HCCA must be added within 30 min.
Note: An inoculated target must be processed within 24 hours of preparation.

5. Run the target on the MALDI Biotyper CA Instrument (see protocol below).

6. If the MALDI Biotyper CA identification of the test organism using the extended direct preparation procedure does not result in high-confidence identification with at least one well with a log(score) value of ≥2.00, **repeat using the Tube Extraction procedure below (yeast) or alternative identification method.**

H. Procedure for Sample Preparation – Tube Extraction

1. Transfer 300 µL of HPLC-grade water into a microcentrifuge tube. Using a sterile 1 µL inoculation loop, inoculate isolated colonies into the water and mix thoroughly until the material is completely in suspension.

2. Add 900 µL ethanol and mix the suspension.

3. Spin down the microbial material in a bench-top centrifuge for two minutes at 13,000- 15,000 rpm. Remove the supernatant.

4. Repeat and remove residual ethanol by pipetting. Air-dry the pellet for at least five minutes at room temperature.

5. Add 25 µL 70% formic acid and pipet the solution up and down until the pellet is resuspended.

6. Add 25 µL of acetonitrile to the tube and mix by pipetting the solution up and down two or three times.

7. Centrifuge the tube for two minutes at 13,000-15,000 rpm.

   Note: Extracted bacteria and yeast suspensions may be held at room temperature for up to 4 h prior to testing on the MALDI Biotyper CA System.

8. Deposit 1 µL each of the supernatant onto **2 sample positions** on a MBT Biotarget 96 US IVD. Add BTS and matrix as described above in the Extended Direct Transfer Protocol (Steps 3-6).

I. Procedure for MBT Sepsityper KIT US IVD

Harvesting blood culture fluid

1. Working inside a Class II biosafety cabinet, disinfect the septum of the blood culture bottle by using an alcohol prep.
2. Insert a vent into the bottle and prepare a smear for Gram stain and inoculate appropriate media.
3. Continue with the extraction if one morphology of bacteria or yeast are seen on Gram stain. Remove the vent and re-disinfect the septum. Using a 3ml syringe with attached needle, remove 1ml of blood from the blood culture bottle and transfer to a 1.5ml centrifuge tube.

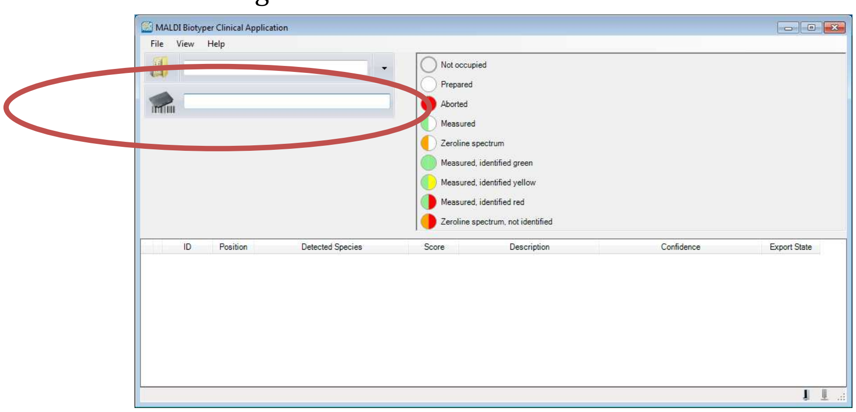
Preparing MBT Sepsityper US IVD samples using the MBT Sepsityper Kit US IVD

4. Add 200 µL Lysis Buffer and mix by vortexing (full speed) for 10 (± 5) seconds.
5. Centrifuge the tube for 2 minutes at 13,000–15,000 rpm at room temperature.
6. Remove the supernatant by pipetting and discard. Add 1 mL Washing Buffer and vortex to resuspend the pellet.
7. Centrifuge the tube for 2 minute at 13,000–15,000 rpm at room temperature.
8. Remove the supernatant from the pellet (in the following 'Sepsityper Pellet'). Add 300 µL HPLC-grade water and resuspend the Sepsityper Pellet by pipetting up and down.
9. Add 900 µL ethanol and mix the suspension with a vortex mixer (full speed) for 10 (± 5) seconds.
10. Centrifuge the tube for 2 minutes at 13,000 - 15,000 rpm at room temperature.

11. Remove the supernatant.
12. Centrifuge the tube for 2 minutes at 13,000 - 15,000 rpm at room temperature.
13. Remove the residual ethanol by pipetting and discard.
14. **Allow the pellet to dry for 5 (±1) minutes at room temperature.**
15. Add 2–50 µL 70% formic acid and thoroughly resuspend the pellet by pipetting up and down. The amount of 70% formic acid added should be proportional to the size of the pellet.
16. Add an equal volume of acetonitrile and mix the suspension by pipetting up and down two to three times.
17. Centrifuge the tube for 2 minutes at 13,000–15,000 rpm at room temperature.
18. Each sample should be run in triplicate by pipetting 1 µL of the supernatant onto an unoccupied MALDI target plate position and drying at room temperature.
19. For validation of the IVD system, inoculate 1µl US IVD BTS onto two target plate positions.
20. Overlay each sample spot and US IVD BTS spots with 1 µL US IVD HCCA solution.
    Note US IVD HCCA solution must be added within 30 minutes after sample spots have dried, otherwise the inoculation must be performed again.
21. Following addition of US IVD BTS and US IVD HCCA the samples of the test microorganism run through the Sepsityper identification process.
22. The final identification must be confirmed from the culture plate by verifying expected organism colony morphology.
23. If an identification of an organism is unclaimed, the target can be re-run on the RUO instrument. Criteria for result reporting must be met as indication is SOP 8 Identification of Bacteria Directly from Blood Culture Bottles by MALDI-TOF (Matrix-Assisted Laser Desorption Ionization Time-of-Flight) Mass Spectrometry**.**

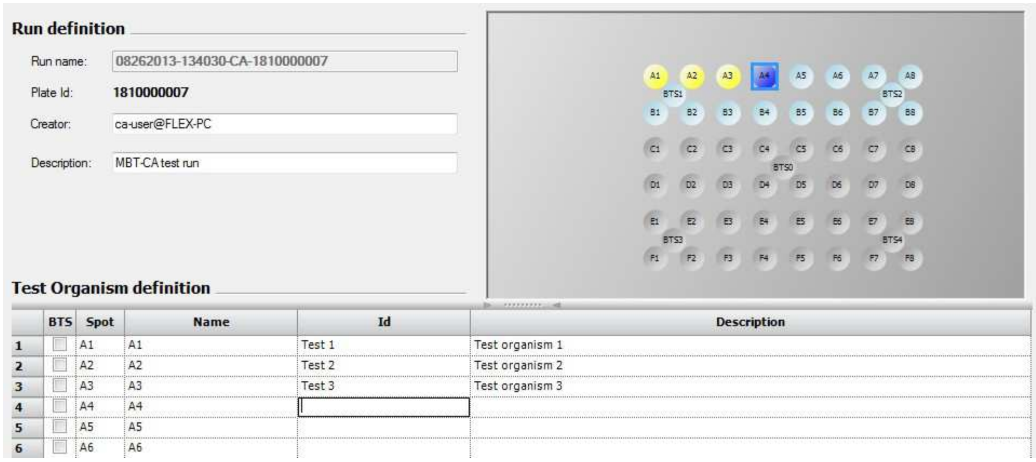
J. Procedure for MALDI Biotyper CA Operation (for more detailed instructions and troubleshooting see the user manual)

1. Start the MALDI Biotyper CA System Software by clicking the MALDI Biotyper CA System Software shortcut on the desktop.
2. After successful start-up, the main application window is displayed. Click on the box next to the target icon and scan the target barcode.

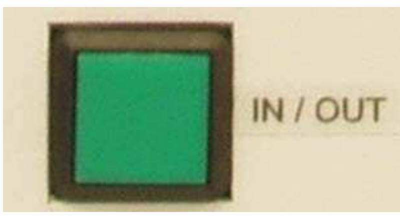


3. Identify the sample positions and BTS control positions from the target to be included in the test run. Highlight target positions by drawing a box and right clicking or selecting individual wells. Choose "add test organisms." Indicate the 2 BTS positions by checking the BTS box to the left. Enter the order number in the ID field by typing or scanning specimen the barcode label. The description field is optional. **When running samples from blood cultures extracted with the Sepsityper kit, Sepsityper must be selected from the dropdown column.**

Note: If Epicenter plate mapping is used, sample ID will be automatically transferred after the target is scanned.



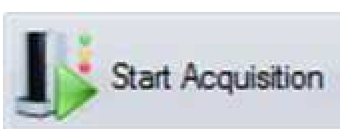
4. Make sure that the target carrier (platform) of the Microflex is in the **OUT** position by pressing the green button on the instrument.



5. Open the load port lid and place the MBT Biotarget 96 US IVD in the MSP Biotarget Adapter into the instrument. Once the target has been loaded, close the lid and press the MALDI target plate **IN/OUT** button once.

   Note: When exchanging targets, do not leave the load port lid open for any longer than necessary. The system must remain closed to prevent delays in establishing vacuum.

6. Once the target is loaded, click Start Acquisition.



7. A PDF result report can be generated and printed at any time by clicking the **View Results Report** button, then print. Results will also be transferred to Epicenter and identifications with a score ≥2.0 will be transferred to the LIS.

J. Workflow

- The FDA-approved, MALDI CA workflow requires that well isolated colonies be tested that are at least 18 hours old.
  - The MALDI CA system **will not** be the primary instrument used for blood culture isolates from plates since we routinely perform MALDI (RUO system) on sweeps from culture plates after less than 18 hours of growth. Well isolated colonies from blood at least 18 hours old can be tested.
- Bacteria that are identified using the MALDI CA as a non-FDA claimed organism (name is reported but greyed out on report) cannot be reported until additional testing is performed.
  - If the organism is of clinical significance, the MALDI target plate can be re-run on the RUO MALDI and reported if the identification meets criteria found in SOP 45 (plates) or 8 (blood culture bottles).
- The MALDI CA system can be used for the identification of yeast from plates and blood culture bottles.

- The MALDI RUO system **will not** be used for yeast identification.

K. Limitations

- The MALDI Biotyper CA System can only be used for the identification of Gram-negative bacteria, Gram-positive bacteria and yeasts included in the reference library.
- Some MBT-CA identifications are non-clinically validated organisms. In the interest of public health, these organisms are displayed but grayed out in the MBT-CA report as a means of directing the required additional laboratory testing. Results for non-clinically validated organisms cannot be transmitted from the MBT-CA to the laboratory information system.
- The log(score) value ranges defined in the MALDI Biotyper CA System represent the probability of organism identification. Results should be reviewed by a trained microbiologist and final organism identification should be based on all relevant information available.
- Do not test directly from a patient specimen (with exclusion of positive blood cultures).
- Testing from plates must be performed using a single isolated colony (or several morphologically similar small colonies if Extraction procedure will be applied).
- A genus only or systematic group identification maybe provided for some organisms, i.e. *Salmonella*.
- If a select agent is suspected, rule out select agents prior to MALDI-TOF MS analysis.
- When using the MBT Sepsityper Kit US IVD, ambiguous results may occur for mixed cultures. In such cases, typically only one microorganism of the mixture is identified, no identification is obtained or a mixed culture hint is generated.
- The general lowering of log(score) levels for all Sepsityper workflows showed a significantly improved performance rate compared to the standard levels [log(score)>1.8 instead of 2.0 for high confidence identification] since lower spectra quality is expected and will result in lower log(scores). As could be expected, such lowering of log(scores) results in a slightly higher error / inaccuracy rate. Such a higher error / inaccuracy rate is the tribute to the improved identification performance and must be seen as a trade-off between higher identification performance and higher error / inaccuracy rate. Altogether no major incorrect results were observed using the standard log(scores) but a large number of identifications resulted in "no identification". From around 1200 spectra analyzed, the log(scores) of 160 spectra turned from red ("no ID") to a low or even high confidence identification without generation of incorrect results.
- In some clinical samples the formation of clots / clumpings (e.g. for *C. glabrata* or *B. fragilis*) can be observed and can lead in rare cases to diminished performance of Sepsityper identification of such samples. Reduced performance (i.e., increased number of no-identifications) is seen with yeast with the MBT Sepsityper as compared to colony testing. This may be due to the lower biomass and protein concentration in yeast Sepsityper pellets.

L. Result Reporting

- Only MALDI Biotyper CA FDA-claimed organisms can be reported. Unvalidated results cannot be reported to the chart.
- At least one BTS well must have an identification score ≥2.0 on the report or the run is invalid.
- An isolate from a **culture plate** must have at least one spot identified from a culture plate with a score **≥2.0** for the species level ID to be reported. For **Sepsityper** samples, one spot with a score of **≥1.8** can be reported to the species level.
- If the duplicate (bacteria) or triplicate (yeast and Sepsityper) give identifications of more than one organism with a score ≥2.0 (ie. S. aureus and S. epidermidis), the ID must be repeated.
- Because the database does not include Shigella, non-lactose fermenting colonies with an identification of E. coli must be confirmed by additional biochemical tests.
- **Reporting of S. pneumoniae or S. mitis/oralis group must be reported as "to be confirmed" unless the full tube extraction procedure, Sepsityper kit, or additional biochemical testing is performed.**
- Identifications of organism "group" or "complex" ie) Enterobacter cloacae complex, should be reported as such and not as "Enterobacter cloacae" unless additional biochemical testing is performed.

M. Quality Control NOTE: SOP contains QC sheets for both FDA-approved Biotyper CA and RUO instrument

- Two spots of BTS are required for each run acquisition to start. At least one spot must pass the calibration and validation or the run will automatically abort.
- At least one BTS score must give an identification of E. coli with a score ≥2.0 in the report.
- Every run must contain a negative control spot that does not contain organism but contains the HCCA matrix. No ID is the expected result.
- S. aureus ATCC 25923 and E. coli 25922 identification must be performed each day of clinical testing using the Extended Direct Transfer protocol. Results should be consistent between the duplicate wells and at least one well must obtain a score ≥2.0.
- Each day of clinical testing of yeast isolates, Candida albicans ATCC 14033, must be included as a control. Results should be consistent between the triplicate wells and at least one well must obtain a score ≥2.0. If a score ≥2.0 is not obtained using the Extended Direct Transfer protocol, the Tube Extraction protocol should be performed.
- If the Tube Extraction Procedure is used for clinical isolates of yeast, then Candida albicans must also be tested the same day using the Tube Extraction Procedure. The expected result is at least one well with a score ≥2.0.
- Each day of clinical testing of Sepsityper specimens, a negative patient blood culture bottle should be spiked with ~10^5 E. coli ATCC strain 25922 and run as the extraction control. At least one spot must be identifies with a score ≥ 1.8.
- **Monthly QC:** Monthly quality control will be performed using a patient negative blood culture and a blood culture containing whole blood and spiked with a patient bacterial isolate when available (*Klebsiella pneumoniae*). If needed, an ATCC strain will be used. Bacterial proteins from both samples will then be extracted using the Sepsityper kit and identified by MALDI. No organisms should be identified from the negative control sample (MALDI score <1.7). For the positive control, the identification of *K. pneumoniae* must be consistent between the three technical replicates and at least one spot must exhibit a score of ≥2.0 (See QC Sheets). We will continue to use the more stringent 2.0 cut-off for species level ID for QC testing.
- **Sepsityper Kit Parallel Testing:** Each new lot or shipment of Sepsityper kits will be tested against the old Sepsityper lot by performing extractions with negative patient blood culture bottles spiked with the QC organisms (*S. aureus* and *E. coli*), and a patient isolate (*K. pneumoniae*) when available. Positive control bottles will be spiked with ~10^5 organisms and monitored in the BD Bactec FX for growth. Negative blood culture bottles will be used as negative controls. Extractions using reagents from old and new Sepsityper kits will be performed as follows:
    1. Old lot lysis buffer and new lot wash buffer.
    2. New lot lysis buffer and old lot wash buffer.
    3. New lot lysis buffer and new lot wash buffer.

    No organisms should be identified in the negative controls (MALDI score <1.7). For the positive controls, the QC organism ID's must be consistent between the three technical replicates and at least one spot must exhibit a score of ≥2.0 (See QC Sheets). The target plate should be run on both the RUO and CA instruments. We will continue to use the more stringent 2.0 cut off for species level ID for QC testing.

N. Instrument Maintenance

- The outside of the instrument must be cleaned daily with 70% alcohol or other disinfectant wipe to disinfect the surface and remove dust.
- The internal O-ring and plate lid must be cleaned at least daily using a gloved finger or with a lint-free tissue (ie. Lens paper or Kimwipe).
- The ion source must be cleaned weekly to remove matrix that builds up on the source. The instrument laser is re-directed at the ion source to burn off any residual material. To clean the ion source, log-in using the RUO username/password, open Flex Control, select "Details" in the lower right window, and click on the first Tab on the top pane "Sub processor System." Under "Ion Source" select "Start Cleaning." The process takes approximately 20 minutes to complete.

O. References

1. MALDI Biotyper 3.1 User Manual Revision K. September 2017. Bruker Daltonics.
2. MBT CA Systems Package Insert Revision G. April 2018. Bruker Daltonics.

3. Patel, R. 2013. Matrix assisted laser desorption ionization time of flight mass spectrometry in clinical microbiology. *Med Micro*. 57(4):564-72.
4. Perry, M. Considerations for Implementation of MALDI-TOF in Public Health and Clinical Microbiology Laboratories. NYSDOH. https://www.aphl.org/conferences/proceedings/Documents/2017/Annual-Meeting/18Perry.pdf. Accessed 12.4.18.